



STANDARD OPERATING PROCEDURE

MX60 Mobile Mapping

Trimble MX60 · Trimble Business Center 2026.10

DOCUMENT	MX60 Mobile Mapping SOP
SYSTEM	Trimble MX60 · Trimble Business Center 2026.10
SET	Technical Manual · Standard Operating Procedure · Field How To · Office How To
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MX60 Mobile Mapping

What Parametrix requires of MX60 mobile mapping work. Short by design: it states requirements and points to the Technical Manual for explanation and to the How To guides for method.

TRIMBLE MX60 · TRIMBLE BUSINESS CENTER 2026.10

Every instruction here says who is telling you. A **Trimble requirement** or an **equipment limit** comes from the manufacturer and is not negotiable on a project. A **recommended practice** is Parametrix's, and a project may have a reason to differ — if it does, that is recorded. See §3.2.

CAUTION Data loss, an irreversible operation, a lost result, or safety	IMPORTANT Gets the job wrong if ignored	TRIMBLE METHOD Trimble documents this — NOT NECESSARILY A REQUIREMENT
TRIMBLE REQUIREMENT · EQUIPMENT LIMIT Binding on the work, on the manufacturer's authority	RECOMMENDED PRACTICE Parametrix recommendation — FOLLOW IT UNLESS THE PROJECT SAYS OTHERWISE	PARAMETRIX REQUIREMENT Required on Parametrix work
SET BY THE PROJECT No single company-wide answer. The project decides, and records it	NO PUBLISHED FIGURE No published number exists. Judgement, and the evidence on the job	

1 Purpose, Scope and Application

1.1 Purpose

This procedure governs the acquisition, processing, quality control, acceptance, delivery and retention of **mobile mapping data collected with the Trimble MX60**.

Its purpose is that a Parametrix mobile mapping deliverable can be relied on, and can account for itself afterwards: what was collected, how it was processed, what it was checked against, who accepted it, and on what basis.

1.2 Scope

This procedure applies to all Parametrix work in which the MX60 is used to collect data that will be delivered, measured from, or relied on — including work where the mobile mapping data is an input to another deliverable rather than the deliverable itself.

It applies from the decision to use mobile mapping on a project **through** archive of the project record.

1.3 What it does not govern

	WHERE IT BELONGS
Conventional survey practice — control networks, datums, adjustment, check observations	Existing Parametrix survey practice. This procedure assumes it
Establishing the control network itself	Existing Parametrix survey practice. §7 states only what mobile mapping additionally requires of it
Terrestrial or airborne scanning	Not this procedure
The client's own specification	Where a client specification is stricter, it governs . Where it is less strict, this procedure governs

The brand guide makes the same point for visual identity: a project deliverable prepared for a client may follow the client's guidelines (*Parametrix Brand Guide*, p.22). The same logic applies to technical requirements.

1.4 Application

This procedure applies to every person performing any activity within its scope, in the roles defined in §4.

Where a requirement cannot be met on a particular project, §22 states what happens. **A requirement is not waived by being inconvenient on the day.**

1.5 Related documents

DOCUMENT	RELATIONSHIP
MX60 Mobile Mapping Technical Manual	The technical basis. Every explanatory reference in this SOP points to it

DOCUMENT	RELATIONSHIP
MX60 Field How To	The field method. Carries the checklists
MX60 Office How To	The processing method in Trimble Business Center
reference/mx60-reference-data.csv	The authority for manufacturer values. Technical Manual, Appendix A
Master register	The single register of open decisions, tests and vendor questions. is this SOP's view of it

The master register is one register. and the Technical Manual's are filtered views of the same file, generated. An item cannot carry one status here and a different status there.

1.6 Review

A Trimble Business Center upgrade is a reason to re-read this procedure. TBC is on an annual release cycle and releases have changed mobile mapping behaviour before. This procedure describes **TBC 2026.10**; where a step depends on a specific dialog or option, the Technical Manual cites the Trimble topic it came from, so a changed interface can be checked rather than guessed at.

2 Related Documents

2.1 The document family

Four documents, used together.

DOCUMENT	ANSWERS
SOP (<i>this document</i>)	What is required, and on whose authority
Technical Manual	Why the system behaves as it does, and what the evidence is
Field How To	How to run the MX60 in the vehicle
Office How To	How to process the data in Trimble Business Center

This SOP governs. The other three support it: the Technical Manual explains the requirements, and the two How To guides give the method for meeting them.

A How To cannot create a requirement. If a How To states something that must be done and this SOP does not require it, one of the two is wrong — and this SOP is the one that decides.

2.2 Where each kind of question is answered

YOUR QUESTION	WHERE
<i>Must I?</i>	This SOP
<i>Why does it work this way?</i>	Technical Manual
<i>What does this term mean?</i>	Technical Manual §6 — the glossary for all four documents
<i>What do I press, in the vehicle?</i>	Field How To
<i>What do I click, in TBC?</i>	Office How To
<i>Where did this number come from?</i>	Technical Manual Appendix B , the reference dataset

2.3 What this SOP does not contain

WHY	
Numerical accuracy tolerances	Trimble publishes no acceptance tolerance for the MX60, and none has been established by test on this system. A tolerance is set by the project (§17)
Explanation	Technical Manual
Step-by-step method	Field How To · Office How To
Terminology	Technical Manual §6. §3 below defines only the terms that carry procedural force here

2.4 Which software this SOP describes

Trimble Business Center 2026.10 and **Trimble Mobile Imaging Rev L**, running on a Trimble MX60.

Trimble changes both. Where a procedure here depends on a specific dialog, option or file name, the Technical Manual cites the Trimble topic it came from, so a changed interface can be checked against the source rather than guessed at.

3 Definitions

Technical Manual §6 is the authoritative glossary for all four documents.

Trajectory, SBET, boresight, lever arm, registration, target, GCP, validation point, Update Scans, Cleanup and the rest are defined there and are not repeated here.

This section defines only the words that carry **procedural force in this document** — the words that decide whether a clause has been complied with.

3.1 Workflow stage names

The nineteen workflow stages are named identically in all four documents and in every checklist and form. They are listed in **Technical Manual §4**.

A stage is not a command. *Registration* is the stage; *Register a Run* is one of three commands that perform it.

3.2 Obligation

TERM	MEANING
shall	A requirement that binds now. Departure is a non-conformance and is handled under §22
shall not	A prohibition that binds now. The same applies
should	A recommendation. Departure is permitted and, where it affects the deliverable, is recorded
may	A permission. No obligation either way

Which clauses bind, and on whose authority

A requirement does not become binding only by Parametrix adopting it. **Trimble's instructions and the equipment's limits bind regardless**, because their authority is the manufacturer's and does not wait for a company decision.

Every clause in this procedure that carries procedural force therefore carries an **authority label**:

LABEL	BINDING NOW?	VERB USED
TRIMBLE REQUIREMENT	Yes	shall · do not
EQUIPMENT LIMIT	Yes	shall · do not
PARAMETRIX REQUIREMENT	Yes — recorded in	shall

LABEL	BINDING NOW?	VERB USED
TRIMBLE METHOD	No — Trimble documents the method without requiring it	should
RECOMMENDED PRACTICE	No — a recommendation from this project	should
SET BY THE PROJECT	No — the answer is not set	<i>no imperative</i>
NO PUBLISHED FIGURE	No — nobody has the answer yet	<i>no imperative, or a stated interim posture</i>

Every clause that tells you to do something therefore says who is telling you.

3.3 Terms with procedural force

TERM	MEANING IN THIS PROCEDURE
Approved	A named person with the authority in §4 has recorded a decision, with a date. An approval that is not recorded did not occur
In writing	Recorded in the project record in a form that survives the project and identifies its author and date. An instant message is not in writing
Recorded	Written into the project record at the time, not reconstructed afterwards
The project record	The durable record of the project, in the location §20 requires. Not a processor's local machine
Independent check	An observation that took no part in any adjustment applied to the data it is checking, and whose designation was fixed before that adjustment was computed (§7.3)
Accepted	The person with acceptance authority under §4 has recorded that the dataset meets the project's stated accuracy requirement. Acceptance is against a requirement, never against a feeling
Delivered	Released outside Parametrix, or relied on by another discipline as final
Destructive operation	An operation that removes data or history and cannot be undone within the software. §18 governs these
Re-collection	Returning to site to collect again. The remedy of last resort, and the only remedy for a field error

3.4 Two words this procedure avoids

"Verified", unqualified	Verified against what, by whom? The procedure names the comparison every time
"QC'd"	§16 defines what a quality control activity consists of. The abbreviation hides whether anything was inspected

4 Roles, Responsibilities and Authorities

4.1 Why this section exists first

Three activities in this workflow change what the deliverable is, and two of them are invisible afterwards:

- **Designating which control points are held as independent checks** — because if the person computing the adjustment also chooses what it is measured against, the check is not independent
- **Accepting a registration** — because acceptance is the point at which the data becomes the deliverable
- **Running a destructive operation** — because it cannot be undone (§18)

TBC makes the first of these a checkbox (*Technical Manual §22.2*), which makes it easy to change quietly. That is the reason this SOP treats authority as a control and not as an organisation chart.

4.2 The decision

SET BY THE PROJECT

Who may operate the system, who may register, who may accept a registration, who may run Cleanup, and who signs an accuracy statement?

Until this is answered, every clause in this procedure that names a role — *the Project Surveyor, the Processor, the System Owner* — is naming something nobody holds. That is why those clauses are **proposed** rather than binding, and why §18 cannot require an authorisation from a person who has not been appointed.

How the duties divide

RECOMMENDED PRACTICE — HOW THE DUTIES DIVIDE

Activity | Performed by | Reviewed or approved by

---|---|---

Mission planning | Processor or Project Surveyor | Project Surveyor

Control network design | Project Surveyor | —

Field acquisition | Field Technician | —

Field quality checks and close-out | Field Technician | —

Trajectory processing | Processor | —

Calibration | Processor | System Owner

Registration | Processor | Project Surveyor

Designating control versus independent check | Project Surveyor | —

Point cloud and imagery QC | Processor | Project Surveyor

Cleanup Mobile Mapping Mission | Processor | Project Surveyor, in writing

Export and delivery | Processor | Project Surveyor

Accuracy statement | Project Surveyor | —

Archive | Processor | Project Manager

The three rows in bold are the ones where the reviewer genuinely matters. The rest could reasonably collapse onto fewer people on a small job.

4.3 The one separation this procedure treats as structural

RECOMMENDED PRACTICE

The person who designates control versus independent check should not be the person who computes the registration.

If both are the same person, the residuals on the check points measure the fit of an adjustment to observations that person was free to choose. That is not an independent check, whatever the numbers say.

On a job small enough that one person does both, the designation is **recorded before the registration is computed** and is not changed afterwards (§7.3).

4.4 Responsibilities that attach to any assignment

Whoever holds a role, these attach to it:

Stand-down authority	The Field Technician may stop or decline collection on safety or data-quality grounds without seeking approval first. The decision and its reason are recorded. This authority is not conditional on being right
Escalation	Anyone who finds that a delivered dataset may not be what it was believed to be escalates it. To whom is part of
Competence	Nobody performs an activity they are not qualified for under §5, whatever the schedule says

4.5 What the record must answer

RECOMMENDED PRACTICE

Whatever Parametrix decides about roles, the project record **should** be able to answer, for any dataset, years later:

1. Who collected it, when, and in what conditions
2. **Which trajectory the delivered data was built on** (*Technical Manual §30 — this one is genuinely hard*)
3. Which points were used as control and which were held as independent checks
4. What the residuals were, on both
5. Who accepted it, on what date, against what accuracy requirement
6. Whether a destructive operation was run, by whom, and what was archived first

Six facts. None is onerous to record at the time. All are effectively unrecoverable later.

4.6 Records this section requires

- Who holds each role on a given project
- Who designated control versus independent check, and when
- Any exercise of stand-down authority, and its reason

5 Competence and Training

5.1 The principle

Mobile mapping fails quietly. A weak trajectory produces a clean, dense, internally consistent point cloud in the wrong place (*Technical Manual §3.2*). There is no visual tell, so the protection is a person who knows what to check and is required to check it.

Competence here is not "has operated the system." It is "knows what the system cannot tell you."

5.2 What qualification covers

SET BY THE PROJECT

Three qualifications are distinguished. Who holds each, how it is obtained and how it is evidenced is part of.

QUALIFICATION	COVERS
Qualified to operate	Installation and pre-flight; initialization and the closing sequence, and why each exists; operating limits and stand-down authority; field quality checks and the field record
Qualified to process and register	The data chain and what regenerates from what; trajectory processing; scan generation; registration and the three commands; what RMS can and cannot prove ; the layered QC in §16
Qualified to accept	All of the above, plus the accuracy framework in §17 and the authority under §4

5.3 The five things a qualified person is expected to know

Not a training syllabus — a list of the misconceptions that have actual consequences. Each is covered in the Technical Manual at the reference given.

	WHERE
1 The trajectory is the job. Every point inherits its error, and an attitude error grows with range	Technical Manual §2, §3
2 A good RMS does not prove success. A bad RMS proves failure. Trimble states this in identical words in two places	Technical Manual §23

		WHERE
3	Registration does not move points. The cloud is unchanged until Update Scans runs	Technical Manual §19
4	A Local adjustment does not extrapolate beyond the outermost control point, and nothing shows where it stopped	Technical Manual §21
5	Cleanup cannot be undone	Technical Manual §28

The In Plain Language boxes in the Technical Manual are the intended route to this. Read end to end with nothing else, they describe the whole workflow in ordinary language.

5.4 Currency

SET BY THE PROJECT

Whether a qualification lapses, and what refreshes it. One trigger is not discretionary: **a TBC release can change mobile mapping behaviour** (§2.3), and a processor working from a prior release's understanding is working from a stale procedure.

5.5 Records this section requires

- Who holds which qualification, and from when
- Training delivered, and against which document

6 Project Setup Requirements

6.1 Before any data is collected

Four things **should** exist in writing before mobilisation. None is onerous, and the failures they prevent are expensive.

RECOMMENDED PRACTICE — THE PRACTICE OF RECORDING THEM. WHAT EACH ONE SAYS IS

a separate open decision, named in the table.

#	REQUIREMENT
1	The accuracy requirement , stated in writing, with the client agreement or scope it derives from
2	The coordinate reference system, datum, epoch and geoid model , stated in writing and matching the control network
3	Grid or ground , agreed with the client in writing
4	The system configuration and fitment the work assumes

6.2 The coordinate reference system

CAUTION

Set the project coordinate system before importing the mission.

Changing it afterwards is possible in TBC generally, but by then every derived product — scans, registrations, exports — was computed in the previous frame. **Treat it as irreversible in practice.**

SET BY THE PROJECT

Which datum and epoch does Parametrix work in, who sets it, and who checks it?

Epoch carries more weight here than in conventional work, because the trajectory's reference frame and the project's control may be realised at different epochs (*Technical Manual §12.4*). TBC 2026.10 allows working at a non-default epoch and warns that the feature "is intended for experienced users, as incorrect settings may lead to inaccurate results" (*TBC RN 2026.10*).

The frame the trajectory is computed in

The trajectory is produced by POSPac, which has its own view of the project's coordinate system and may compute in ITRF00 and then transform. The only outward sign is the SBET filename (*Technical Manual §12.3, §17.4*).

SET BY THE PROJECT

IN-Fusion+ Single Base or PP-RTX? The two differ in whether a local base station must be occupied on every mission, and in the reference frame the solution is computed in. It interacts with (§8) and with above.

NO PUBLISHED FIGURE

Which of Parametrix's normal coordinate systems POSPac recognises directly, and which trigger the ITRF00 path. Answerable once, then known.

6.3 Grid and ground

IMPORTANT

A **ground-scaled** export **does not record the scale factor it used**. A **grid** export writes a sidecar naming the coordinate system and scale factor
TBC 11769; Technical Manual §12.5.

The recipient of a ground-scaled file cannot recover the scale factor from the file. Agree it in writing, and make sure the delivery can say what it is (§19).

SET BY THE PROJECT

Deliverable specification — standard formats, which export path produces each, and the default scaling.

6.4 The system this work assumes

The **Parametrix system is the MX60 Premium** — the top configuration. Panoramic imagery is **12288 × 6144 px**, roll and pitch accuracy **0.0025°**, and the better of the two published 60-second outage figures applies (*Technical Manual §7.3, §15, §16*).

SET BY THE PROJECT — · /

Are GAMS and DMI fitted, and which mounting rack is on the vehicle?

Three facts, one phone call. GAMS decides whether the initialization manoeuvres are required or merely advisable; DMI decides whether the published no-outage accuracy applies to this system at all and whether DMI settings appear in trajectory processing; the rack decides whether the published GAMS corner offsets apply (*Technical Manual §7.7, §9*). **The vendor can confirm all three from the serial number**, along with the configuration itself, which has not yet been checked against it.

6.5 Records this section requires

RECORD	WHERE
Accuracy requirement, and its source	Project record
CRS, datum, epoch, geoid — and who set them	Project record
Grid or ground, as agreed	Project record and client agreement
Configuration assumed — Premium — and the fitment assumed with it	Project record

7 Control Requirements

This section states only what **mobile mapping** additionally requires of a control network. Establishing the network is ordinary Parametrix survey practice and is not governed here (§1.3).

7.1 Control has to be findable in a point cloud

A mobile mapping control point must be identifiable in the point cloud at the density and incidence angle the vehicle produced. **That is a different requirement from occupiable with a prism** (*Technical Manual §22.5*).

WORKS	WORKS POORLY
Painted stop-bar and lane-line corners — crisp intensity edges	A survey nail in asphalt: a few millimetres across, below cloud resolution
Target panels for which TBC holds templates	Small features at grazing incidence
Painted arrow and legend corners	Anything that moves, is repainted, or is worn

Horizontal and vertical suitability are separate. A painted stop-bar corner is an excellent horizontal target and a poor vertical one, and TBC allows a point to participate in one component and not the other (*Technical Manual §22.2, §22.3*).

NO PUBLISHED FIGURE

Which feature types are fit for horizontal control, vertical control, or both, at MX60 point density and incidence angle. **This will shape control design more than any software setting.**

7.2 Control and the delivered extent

CAUTION

Local does not extrapolate. Trimble states it is *"not for systematic error along the run or for adjusting outside the ground control points set"* (TBC 22905).

Beyond the first and last control point the trajectory is not adjusted, **and nothing indicates where the adjustment stopped**. Control must bracket the extent you intend to deliver.

TRIMBLE REQUIREMENT — *binding now, on Trimble's authority, not Parametrix's*

The limitation itself is Trimble's and is not open to local interpretation. Data outside the outermost control point **shall not** be described as registered to that control, because it was not adjusted.

RECOMMENDED PRACTICE

What follows from the limitation is a design rule, and the design rule is ours: control **should** bracket the extent to be delivered.

Trimble states a limitation, not a control-design requirement. It does not say control must bracket anything; it says the adjustment stops. This procedure draws the practical consequence and proposes it — it does not present it as a manufacturer instruction.

Bracketing, not merely falling within. The ends of a corridor are also where the trajectory smoother had data on one side only (*Technical Manual §14.4*), which makes them simultaneously the weakest data and the place registration helps least.

7.3 Independent checks are designated before registration

RECOMMENDED PRACTICE

1. **Independent check points are designated by the Project Surveyor before registration begins**
2. **A point's As Check status is not changed during processing.** If a designation was wrong, it is changed by the Project Surveyor, **recorded**, and the registration is recomputed from the imported trajectory using **Edit** — not layered on top of the previous one (*Technical Manual §21.8*)
3. **The designation is recorded in the project record** and travels with the accuracy statement

The failure this prevents. A conscientious processor registers a mission, finds one check point with a residual larger than expected, and adds it to the adjustment to bring it in. Every step is well intentioned. The result is an adjustment with no independent check at all, and a set of residuals that now measure nothing (*Technical Manual §22.4*).

7.4 How much control, and where

SET BY THE PROJECT

How many control points, at what spacing, how many held as independent checks, and does density vary with predicted GNSS conditions?

No Trimble source states a minimum, a spacing or a ratio. TBC's software minimum is one control pair — a mathematical floor with no bearing on survey adequacy (*Technical Manual §21.6*).

RECOMMENDED PRACTICE — *a structure for the decision, not the decision*

- Control **bracketing** each end of the delivered extent, outside it where the corridor allows
- Control along the corridor at an interval set from the project accuracy requirement, and **tightened where GNSS is predicted to be degraded** (§8)
- **Independent check points distributed, not clustered**, including at least one in each distinct GNSS environment — open sky, tree cover, urban canyon
- **A check point near each end of the delivered extent**

No counts, spacings or ratios appear here deliberately. Those are the content of.

Two indications of scale exist, and neither is a specification. Trimble's Target-Bundle Adjustment operates at **250 m** intervals when checked and **70 m** when unchecked (*Technical Manual §21.7*). The Queensland TMR mobile laser scanning guideline is an example of how another agency answered this question, cited in the Technical Manual §22.6 **as an example and not as a standard**.

7.5 Records this section requires

- Control network, with coordinates and their source
- **Which points are control and which are independent checks, fixed before registration**
- Who designated them, and when

8 Mission Planning Requirements

8.1 What planning must produce

A mission plan, in the project record, before mobilisation. It states the route, the passes, the GNSS assessment, the initialization locations, and — explicitly — **what mobile mapping will not get**.

8.2 Passes and overlap

SET BY THE PROJECT

How many passes, in what pattern, by roadway type?

No Trimble source specifies a pass pattern. The v1 Parametrix draft stated a minimum of three passes on the pavement; **that figure is not traceable to any source in the evidence set** and is recorded here so it is not lost, not because it is established.

Two facts constrain the answer, and both are in the Technical Manual:

- **A second pass in the opposite direction improves a dataset more than any setting change**, because it converts grazing incidence into direct incidence and long range into short range for the far side of the corridor (*Technical Manual §16.3*)
- **Overlap is what makes the office remedies possible**. LiDAR QC and run-to-run registration both require overlapping runs (*Technical Manual §11.5, §21.11*)

CAUTION

If a GNSS-hostile stretch planned for two passes receives one, **both LiDAR QC and run-to-run registration become unavailable** for that stretch. There is no software warning, and no way to tell afterwards except by looking at what is there.

8.3 GNSS assessment

RECOMMENDED PRACTICE

The plan **should** identify GNSS-hostile stretches **before mobilising**, and for each state the expected duration at realistic collection speed.

Duration, not length. Inertial drift is a function of time. A 300 m tunnel at 80 km/h is 13 seconds; the same tunnel at 20 km/h in traffic is nearly a minute (*Technical Manual §15.1*). The time of day is part of the assessment.

IMPORTANT

Trimble publishes positioning performance at **no outage** and after a **60-second outage**, and nothing beyond MX60 UG Rev B, p.56; Technical Manual §15.2.

Outages materially longer than 60 seconds are a planning problem, not a driving problem. Beyond the published figure, any expectation is an extrapolation past the manufacturer's stated envelope.

For each hostile stretch the plan states the mitigation: additional control, planned overlap for LiDAR QC, or a different acquisition method — **because two of the three have to be arranged before the crew leaves.**

NO PUBLISHED FIGURE

Whether predicted GNSS conditions correlate with achieved trajectory RMS on this system. Until that is established, the assessment is judgement rather than estimate.

8.4 Initialization locations

RECOMMENDED PRACTICE

The plan identifies a **primary and a backup** initialization location, each of which is:

- open sky, away from buildings and canopy
- somewhere the vehicle can safely sit still for two to three minutes
- with room afterwards to drive straight and perform dynamic manoeuvres

Scout them on imagery before mobilising. A lot that turns out to be fenced, occupied or under trees costs twenty minutes at the worst moment of the day. (*Technical Manual §13*)

8.5 Base station strategy

SET BY THE PROJECT

Own base on project control, VRS, RTX, or CORS post-processing — and what maximum baseline?

It interacts with (§6.2): Single Base requires a local base station on every mission, PP-RTX does not. The decision determines field logistics on every job.

8.6 What mobile mapping will not get

RECOMMENDED PRACTICE

Record in the project file **before mobilising**: the segments where mobile mapping is expected to be marginal, the mitigation chosen for each, and the segments where another method is proposed.

This is the clause people avoid, and the reason it is a requirement rather than a recommendation. What it costs to discover the same thing in the office instead is set out in **Technical Manual §15.5**.

SET BY THE PROJECT

The decision rule when a corridor produces an unacceptable trajectory: who decides, against what, and what the client is told.

8.7 Records this section requires

- The mission plan — route, passes, direction, overlap
- GNSS assessment, with duration estimates and mitigations
- Initialization locations, primary and backup
- Segments mobile mapping will not serve, and what is proposed instead

9 Field Acquisition Requirements

The method is in the **Field How To**. This section states what shall be done, and what shall not.

9.1 Before the vehicle moves

#	REQUIREMENT
1	The installation configuration, lever arms and Vehicle Preset are as recorded, and have not changed since
2	Where GAMS or DMI are fitted, each is activated in Vehicle Settings — not merely installed
3	The mission plan (§8) is on board and understood
4	Storage has capacity for the planned collection

CAUTION

An aiding sensor that is installed and wired but **not activated in Vehicle Settings logs nothing**, and nothing looks wrong

TMI UG Rev L, p.21; Technical Manual §9.3. The office symptom appears days later as a dimmed settings pane, by which time the mission is collected.

SET BY THE PROJECT

Where are the lever arms, the Vehicle Preset and the installation configuration recorded, and who verifies them? These values are entered once and used on every mission afterwards. An error in them is systematic, invisible, and persists until somebody re-measures.

9.2 Initialization

TRIMBLE REQUIREMENT — MX60 QSG REV B, §5.3, P.11

Navigation alignment shall be complete before data logging begins. Trimble states it in mandatory terms and the system enforces it: *"Navigation alignment must be done first before data logging is allowed!"* It is not a matter of operator discipline.

TRIMBLE REQUIREMENT — MX60 QSG Rev B, §5.4, p.13

A mission shall be at least 30 minutes long. *"Important! A minimum mission time of ≥30 min is required."*

TRIMBLE METHOD — MX60 QSG Rev B, §5.3, p.11; §6, p.14

Trimble documents an initialization sequence — static logging, a straight run, then dynamic manoeuvres — and it **should** be performed at the start of every mission, in the order given.

Trimble does not state this sequence as a requirement, and this procedure does not claim it is one. Trimble writes *"Mobile Mapping Mission **should** be started in a static mode"*, and heads the in-field list *"Proposal of a checklist for system operation"*. Whether Parametrix makes the sequence mandatory is.

IMPORTANT

Green is not finished. The navigation status turning green means the solution met its accuracy thresholds, not that it has converged. Trimble asks for **up to ten further minutes** before recording anything that matters (MX60 QSG Rev B).

The first data after the light turns green is the weakest data of the day.

RECOMMENDED PRACTICE

That data **should not** be spent on the most important part of the corridor. Trimble asks for the settling time; how the crew spends it is Parametrix's to decide.

The sequence and its rationale are in **Technical Manual §13**; the steps are in the **Field How To**.

9.3 The closing sequence

TRIMBLE METHOD — MX60 QSG REV B, §5.5, P.13

Trimble documents a closing sequence and instructs that the mission be finalized *"according to the following sequence"* — dynamic manoeuvres, varying speed, then 2–3 minutes static — and it **should** be performed at the end of every mission, before the mission is closed in TMI.

Trimble states the reason, not an obligation. Its note says symmetrical start and end procedures *"supports forward and reverse processing modes in the office software"*. That is a strong technical reason and a weak instruction; the strength of the rule is Parametrix's to set, and it is.

CAUTION

Navigation logging stops the instant the mission is closed. Closing the mission and then driving to open sky achieves nothing. (*Technical Manual §14.2*)

It takes about five minutes and it is the cheapest quality improvement in the workflow. It cannot be added later — a crew returning the next day cannot append it, because a new mission is a new trajectory (*Technical Manual §14.3*).

9.4 Operating limits

LIMIT	VALUE	SOURCE
Recommended maximum speed with the system operating	80 km/h (50 mph)	MX60 UG Rev B
Absolute maximum, operating or not	110 km/h (68 mph)	MX60 UG Rev B
Direct sun, stationary or below 10 km/h	Outside the rated operating envelope	MX60 UG Rev B, p.53

CAUTION

Battery Protect MX60 UG Rev B, p.27: an audible warning below **10.5 V** for longer than **12 seconds**, power cut below **10.5 V** for more than **90 seconds**, recovery if voltage rises above **12.0 V** within that time.

An interrupted run loses the closing sequence with it. Treat the audible warning as an instruction to restore charge, not as information.

SET BY THE PROJECT

Field operating rules — wet-weather go/no-go with operator stand-down authority, night collection, and collection speed by deliverable type. Trimble publishes a recommended maximum and an absolute maximum and **no guidance relating speed to deliverable quality** (*Technical Manual §16.2*).

9.5 Stand-down authority

The Field Technician may stop or decline collection on safety or data-quality grounds without seeking approval first. The decision and its reason are recorded. This authority is not conditional on the decision later proving correct (§4.4).

9.6 The field record

RECOMMENDED PRACTICE

Recorded per mission, at the time: date, operator, vehicle, mission ID; capture settings; the initialization location and time; each run with start and end and any incident; **GNSS conditions observed**; weather; traffic and occlusion events; **anything not collected and why**; the closing sequence performed; disk and free space at the end.

This is the one record in the whole workflow with no software artefact behind it. Nothing in TBC knows that a truck occluded the near lane for 200 m, or that the corridor was collected in rain. If the crew does not write it down, it is gone (*Technical Manual §30*).

9.7 Records this section requires

- Pre-flight confirmation, including aiding-sensor activation where fitted
- The field record, per §9.6
- Any exercise of stand-down authority, and its reason
- Operating-limit exceedance, if any, and what was done

10 Field Close-out and Handoff

10.1 Why close-out is a separate requirement

Everything in this section is cheap on site and impossible afterwards. A gap discovered in the office costs a mobilisation; the same gap discovered before the vehicle leaves costs twenty minutes.

The field stage is the only stage in the workflow with no office remedy
(*Technical Manual §5.5*).

10.2 Coverage verification, before leaving site

RECOMMENDED PRACTICE

Confirmed against the mission plan (§8) before the vehicle leaves:

1. Every planned pass was driven, **in the planned direction**
2. The corridor was driven end to end, including any extent beyond the deliverable needed to bracket control (§7.2)
3. **Planned overlap was actually collected** — this is the one that removes office options if missed (§8.2)
4. Sections not collected, and why, are recorded
5. The closing sequence was performed (§9.3)

10.3 What accompanies the data

RECOMMENDED PRACTICE

A mission is handed to the office with:

|

---|---

The complete mission folder | Not the `.mxd` alone — that is an index, not the data (*Technical Manual §5.2*)

The field record | §9.6

Base station data | If a local base was occupied

Any deviation from the plan, and its reason | §8, §22

10.4 Handoff is a transfer of responsibility

Until the office has confirmed a verified copy (§11), **the field holds the only copy of an unrepeatable measurement.** Close-out is not complete when the vehicle is packed; it is complete when §11.2 has been satisfied.

10.5 Records this section requires

- Coverage verification, performed and by whom
- Deviations from the mission plan
- Handoff — what was transferred, to whom, when

11 Data Transfer and Custody

11.1 The one irreplaceable thing

`POS_1/raw/` holds the raw GNSS and inertial observations. **Every trajectory the office ever computes derives from those files, and nothing can recompute them** (*Technical Manual §5.2*). The picked registration targets are the other irreplaceable item, and that comes later (§13).

Everything else in the chain is reproducible. These two are not.

11.2 Transfer and verification

RECOMMENDED PRACTICE

1. **Copy, do not move.** The source disk remains the source until a verified copy exists in two locations
2. **Verify the copy** — file count and total size at minimum, a checksum comparison where the tooling allows
3. **Confirm the `.mxd` opens.** Importing into a scratch TBC project is the definitive test
4. **Confirm `POS_1/raw/` is present and non-empty.** Without it there is no post-processed trajectory and the mission is NAV-only (*Technical Manual §17.2*)
5. **Confirm base station data** is present if a local base was occupied
6. **Only then** is the source disk available for reuse

CAUTION

Do not clear a data disk until a verified copy exists in at least two locations and the `.mxd` has been confirmed to open.

A cleared disk is not recoverable, and a mobile mapping mission is not re-drivable at reasonable cost.

11.3 Backup before processing

RECOMMENDED PRACTICE

The raw-data backup is taken before any processing begins, not after.

Processing writes into the project and, with **Backup SBET Next to MXDB** enabled (§13.2), into the raw data folder alongside the `.mxd`. A backup taken after processing has begun is a backup of a partly processed state — usually fine, and occasionally exactly the wrong thing to have.

11.4 Structure, naming and location

SET BY THE PROJECT

Folder structure, naming convention and storage location.

RECOMMENDED PRACTICE

Two constraints are proposed as non-discretionary whatever else decides: the project record **should not** live on a processor's local machine (§3.3), and raw mission data **should** be distinguishable from processed products without opening them.

11.5 Chain of custody

SET BY THE PROJECT

Is a chain-of-custody record required? For most survey work the answer is probably no. For work that may be used in a dispute, a claim or a proceeding, it is a different question, and it has to be decided **before** the data is collected rather than when it is asked for.

11.6 Records this section requires

- Transfer performed, verified how, by whom, when
- Location of the raw-data backup
- Custody, where required

12 Office Intake Requirements

12.1 The principle

Everything checked at intake is cheap; everything discovered later is not. Intake is also the last point at which re-collection is still a small decision.

12.2 What is verified at import

RECOMMENDED PRACTICE

Before any processing:

| Check | Why

---|---|---

- 1 | **Project coordinate system** matches the control network and the client requirement | Changing it afterwards is irreversible in practice (§6.2)
- 2 | **Covered distance** is consistent with the field record |
- 3 | **Run count** matches the field record | Fewer runs than the crew logged means data was lost in transfer
- 4 | **The active trajectory** is the intended one, and is **SBET rather than NAV** unless there is a recorded reason | (*Technical Manual §17.2*)
- 5 | **Capture Devices** lists the sensors expected for the configuration | A missing camera or laser means a sensor was disabled or failed in the field (§9.1)
- 6 | **Base station data** is present if the trajectory will be processed in house |
- 7 | **The calibration state the mission was collected under is recorded** | See §12.3

Seven checks, none taking more than a minute.

CAUTION

Set the project coordinate system before importing the mission.

Changing it afterwards is possible in TBC generally, but by then every derived product — scans, registrations, exports — was computed in the previous frame.

Treat it as irreversible in practice.

12.3 Capture the calibration state

`Extcal.json` travels with the raw mission data, and the **Mission Report** carries per-sensor boresight and lever-arm calibration **with a date of calibration** (*TBC 24868*).

That dated calibration record is the only one found anywhere in the workflow (*Technical Manual §30*). It is captured at intake because a later Cleanup can remove the objects that would have produced it (§18).

12.4 What intake does not do

RECOMMENDED PRACTICE

Intake should not correct anything. If a check fails, it is recorded and raised (§22).

A processor who quietly fixes a coordinate system mismatch at intake has removed the evidence that the field and office disagreed — which is the reason the practice is proposed, and the reason it matters more than it looks.

12.5 Records this section requires

- Intake checks performed, by whom, with the result of each
- Calibration state at collection — `Extcal.json` and the Mission Report
- Any intake check that failed, and what was done

13 Processing Requirements

What shall be done, not how. The method is in the Office How To; the explanation is in the Technical Manual §§17–21.

13.1 Trajectory

#	REQUIREMENT
1	Survey deliverables are produced from a post-processed SBET , not the real-time NAV solution. Where NAV is used, the reason is recorded and the deliverable is qualified
2	The antenna model is confirmed before computing. The MX60 requires Trimble 112735
3	The frame and epoch the trajectory was computed in is recorded

IMPORTANT

The antenna model is read automatically from the RINEX, which means it can be read **wrongly**. An incorrect model introduces a **systematic antenna-height and reference error** into the trajectory TBC 25943; Technical Manual §17.3. It is one field nobody looks at, upstream of everything.

SET BY THE PROJECT

Does Parametrix hold a POSPac MMS 8.6+ licence, and where is it installed? Without it, trajectory processing inside TBC and the PFI route are both unavailable, which removes one of the three degraded-GNSS remedies (*Technical Manual §10.3, §27*).

SET BY THE PROJECT

Computation mode — Single Base or PP-RTX. Stated at §6.2 because it is a project-setup decision as much as a processing one.

SET BY THE PROJECT

Is LiDAR QC a capability Parametrix intends to have? It is the only degraded-GNSS remedy needing neither a POSPac licence nor additional control, and it needs a workstation with 128 GB of RAM at minimum (*Technical Manual §11.1*). The answer is a procurement decision, not a processing one.

NO PUBLISHED FIGURE

Three trajectory-processing defaults are untested against Parametrix conditions: the **Multipath** default of Medium on open-sky corridors, the **DMI scale factor standard deviation** default of 5 % where the wheel may never have been measured, and the **LiDAR QC** range and laser defaults. Trimble's own guidance beside the LiDAR QC laser setting contradicts its default (*Technical Manual §11.3*).

13.2 Preserve what records the trajectory

RECOMMENDED PRACTICE

Enable Backup SBET Next to MXDB. That log is the only artefact found anywhere in the workflow that records the frame and epoch a trajectory was computed in, and it lives with the raw data rather than inside a TBC project that may later be cleaned up (§18) or lost.

13.3 Scan generation

#	REQUIREMENT
1	The filters applied are recorded. Filter choice is a defensible-or-not decision a reviewer may need to see years later
2	Colorization is applied or not applied by decision, not by default

RECOMMENDED PRACTICE

Capture the Results of Scan Generation into the project record. It is the only artefact that states which filters produced a given cloud.

NO PUBLISHED FIGURE

Filter defaults are untested. **in particular:** whether the **Reflective Panels** filter removes legitimate retro-reflective returns from signs and line marking — on sign and retroreflectivity work, those returns are the deliverable (*Technical Manual §18.3*).

13.4 Registration

Registration has its own section: §14. It is not a processing step like the others — it is the step that decides whether the deliverable sits where it is supposed to, and it carries its own authority, designation, command-selection and record requirements.

13.5 Records this section requires

- Trajectory processing settings, and the frame and epoch log
- Results of Scan Generation

Registration records are §14.10.

14 Registration Requirements

Registration is the step that decides whether the deliverable sits where it is supposed to. It is also the step with the most ways to produce a confident, defensible-looking, wrong result — which is why it has a section of its own rather than a subsection of processing.

This section states what is required. The three commands and their click sequences are the **Office How To §§16–21**; why registration behaves as it does is **Technical Manual §21**.

14.1 When registration is required

- A dataset that will be **measured from, delivered, or relied on** is registered to surveyed control
- A dataset used only to look at, internally, may not need it

RECOMMENDED PRACTICE

Registration to surveyed control **should** be performed on every dataset for which an accuracy statement will be issued. A trajectory that has never been fitted to control has no independent evidence of its absolute position at all — the processing was internally consistent and nothing more (*Technical Manual §8.5*).

14.2 Authority

SET BY THE PROJECT

Who may perform a registration, and who may accept one?

Recommended at §4.2: performed by the Processor, reviewed by the Project Surveyor.

RECOMMENDED PRACTICE

The person who designates control versus independent check should not be the person who computes the registration (§4.3). Where one person must do both on a small job, the designation is recorded **before** the registration is computed, and is not changed afterwards.

14.3 Control and independent check designation

The requirement is §7.3 and is not restated here. Two consequences belong to registration:

- The designation is **fixed before** registration begins
- A point's **As Check** state is **not changed during** processing. If a designation was wrong, it is changed by the person with the authority, recorded, and the registration recomputed from the imported trajectory

TRIMBLE REQUIREMENT — AN ENFORCED SOFTWARE LIMIT

At least one control point in a registration shall not be a validation point.

Trimble states that if every selected GCP is set **As Check**, *"an error will pop-up and will prompt you to have at least one ground control point (GCP) for the calculation"* (TBC 22905).

It is worth knowing which way this cuts: TBC protects the *calculation*, not the *check*. It will stop you holding nothing back for the adjustment. **It will not stop you holding nothing back for the check**, which is the failure below.

The failure this prevents. A processor registers, finds one check point with a larger residual than expected, and adds it to the adjustment to bring it in. Every step is well intentioned. The result is an adjustment with no independent check at all, and a set of residuals that now measure nothing (*Technical Manual §22.4*).

14.4 Command selection

Three commands perform registration and they are not interchangeable (*Technical Manual §21*).

COMMAND	USES SURVEYED CONTROL?	SCOPE
Register a Run	Yes	One run
Register a Mission	Yes — each GCP reusable across runs	A set of runs
Register Run to Run	No — cloud-to-cloud against a fixed Reference Run	A pair, batched

SET BY THE PROJECT

Is Register a Mission the corridor default, with Register a Run reserved for single-run cases and for repairing one run in an otherwise accepted mission?
And where does run-to-run sit?

The decision also has to say what happens to a mission registration when one run is later re-collected.

NO PUBLISHED FIGURE

Which registration **type** — Global, Local, Global-then-Local — and when; whether Target-Bundle Adjustment should be checked; and how much run overlap run-to-run actually needs. No selection rule is published for any of the three.

14.5 Use of surveyed control

TRIMBLE REQUIREMENT — BINDING NOW, ON TRIMBLE'S AUTHORITY

A **Local** registration does not adjust beyond the outermost control point, and nothing indicates where the adjustment stopped. Data outside that bracket **shall not** be described as registered to the control (§7.2).

RECOMMENDED PRACTICE

Control **should** bracket the extent to be delivered. The limitation is Trimble's; the control design rule that follows from it is this project's proposal, not a manufacturer instruction (§7.2).

TRIMBLE REQUIREMENT — *an enforced software limit, not advice*

A GCP and its picked target **shall not** be more than **30 m** apart. TBC rejects the pair beyond that distance — the limit cannot be exceeded, only worked around (*TBC 22905*).

14.6 Run-to-run — what it cannot do

RECOMMENDED PRACTICE

Run-to-run registration improves **relative** agreement between passes. It uses no surveyed control and cannot establish absolute position. It **should** therefore be used only in this order:

1. Register to surveyed control first
2. Assess against independent check points and visually (§16)
3. **Only then**, if overlapping passes still disagree, use run-to-run — choosing as **Reference Run** the pass with the better GNSS conditions and the better residuals against control
4. **Re-check against the independent check points afterwards**, because the Run to Adjust has moved

Step 4 is the one that gets skipped, and it is why the sequence matters: adjusting one run to match another changes its residuals against control, and if the Reference Run was itself displaced, run-to-run propagates that displacement faithfully into the run you adjusted.

14.7 Re-registration, Edit and Reset

CAUTION

Getting this wrong stacks adjustments on adjustments. A processor who registers, dislikes the residuals, and registers again has applied a second correction on top of the first. The residuals will look better. The trajectory has been bent twice against the same control.

To improve a registration, use Edit. To start over, edit and Reset.

RECOMMENDED PRACTICE

A registration that needs changing **should** be recomputed from the **imported** trajectory using **Edit**, never layered on a previous result.

CAUTION

Picked targets are written to `Targets.csv` when Registration Auto-Saving is on. Reopening the command prompts to reload them, and *"if you choose 'No', they will be emptied from the Targets.csv file and you will not be able to retrieve them"* (TBC 22905).

`Targets.csv` holds the registration's observations. **Answering "No" discards the field book.**

NO PUBLISHED FIGURE — whether Registration Auto-Saving is on by default.

14.8 Update Scans

CAUTION

Registration does not modify the point cloud until Update Scans is performed.

An operator can complete a registration, obtain good residuals, accept the result, and then export point cloud data that still reflects the pre-registration trajectory. **The export succeeds. The file is valid. The data is unregistered.**

RECOMMENDED PRACTICE

Update Scans should be run before a registration result is inspected, accepted or exported. The confirmation that it was run is the export gate at §19.2.

14.9 Review after registration

#	
1	Residuals on independent check points are read and recorded, by component
2	The visual check is performed against the registered cloud, not the unregistered one
3	Where run-to-run was used, check points are re-read after it

TRIMBLE REQUIREMENT

"Good RMS values do not mean that the calibration succeeded. A visual check is needed. On the other side, bad RMS values mean that the calibration failed." (TBC 24886, 25096)

A registration **shall not** be judged on its residuals alone. The visual check is Trimble's instruction and does not wait on a Parametrix decision.

14.10 Records this section requires

- Registration name, type, and the runs included
- The trajectory node produced, and its SBET filename **with its** `_reg_####` **number**
- **Control and check designation, with the residual on each point, by component**
- Confirmation that **Update Scans** was run
- `Targets.csv` , archived
- That the visual check was performed, by whom, over what extent

14.11 Acceptance of a registration

A registration is **accepted** under §17, against the project's stated accuracy requirement, by the person with the authority under §4. **Acceptance is not the registrant's** (§17.5).

SET BY THE PROJECT

What constitutes an acceptable registration is not established (§17.2).

Until it is, acceptance rests on documented professional judgement supported by the evidence above — and the decision at §17.2 is whether that is permitted at all.

15 Calibration Control

15.1 What calibration is, in one line

TBC's laser scanner calibration estimates the **angular** offsets between sensors. Lever arms are **measured, not estimated** (*Technical Manual §7.6, §20.2*). A wrong lever arm cannot be calibrated out, because the adjustment has no parameter for it.

An angular error acts through range: the same error is ten times larger at 100 m than at 10 m (*Technical Manual §3.1*).

15.2 Currency

SET BY THE PROJECT

What is the recalibration interval, and what triggers a recalibration outside it?

The specific question that has to be answered explicitly: **does daily removal and refitting of the Sensor Unit count as disturbing the calibration?** If the unit comes off the vehicle between jobs, the answer determines whether calibration is an annual event or a per-mobilisation one.

RECOMMENDED PRACTICE

No mission should be processed against a calibration whose currency cannot be established.

The calibration state in force is captured at intake (§12.3), which is what makes this checkable. The proposal is not controversial; what makes it undecided is that **the interval it would be checked against does not exist yet**.

15.3 The calibration site

SET BY THE PROJECT

Where is the calibration site, and who maintains it?

Trimble specifies the geometry, and two different procedures need compatible sites:

	REQUIREMENT	SOURCE
Laser scanner calibration	Two streets crossing at $90^\circ \pm 30^\circ$, at least 20 m usable on each side and ideally 80 m total, façades in each direction, little vegetation	TBC 24886
LiDAR QC	Two perpendicular strips, each 250–300 m , each driven in both directions; structured scene; open sky	TBC 28972

RECOMMENDED PRACTICE

Establish one site that satisfies both — two streets crossing near 90° , façades on all four approaches, little vegetation, **125–150 m of usable street on each arm**, open sky, drivable in both directions without traffic-control complications.

The two requirements are compatible and the stricter one should govern. Establishing such a site is real work — reconnaissance, a traffic plan, possibly permission — and doing it once, well, before it is needed under schedule pressure is worth more than the procedure it supports.

15.4 Judging a calibration

CAUTION

Trimble states, in identical words in two separate topics:

"Good RMS values do not mean that the calibration succeeded. A visual check is needed. On the other side, bad RMS values mean that the calibration failed." (TBC 24886, 25096)

TRIMBLE REQUIREMENT — *binding now*. Trimble does not recommend the visual check; it states that one **is needed**, in identical words in two topics.

A calibration shall not be accepted on RMS alone. The visual check is part of the acceptance, not an optional extra (*Technical Manual §23.1, §25*).

15.5 The calibration record

RECOMMENDED PRACTICE

Export the calibration JSON after every calibration and archive it outside the TBC project, named with the system serial number and the calibration date.

It is the complete calibration state of the system in one small file, it can be imported into any subsequent project, and it is the only portable record of what the system's angles were on a given date. Cleanup (§18) or a lost workstation should not take it with them.

15.6 Periodic system verification

Distinct from per-project QC: the check that the **instrument** is still performing.

TRIMBLE METHOD — MX60 UG REV B, P.7

Trimble publishes a target-scanning check of the instrument against total-station truth, and recommends it "regularly" and "especially before starting an extensive data acquisition campaign" — but **gives no interval**. The method is set out in **Technical Manual §31**.

SET BY THE PROJECT

Is this the periodic verification Parametrix uses, and at what interval?

NO PUBLISHED FIGURE

Whether Trimble or the dealer expects this check specifically, and at what period.

15.7 Records this section requires

- Calibration performed — date, site, who, and the result including the visual check
- The calibration JSON, archived outside the project
- Which calibration each mission was processed against
- Periodic verification, when performed

16 Quality Control Requirements

16.1 The principle this section rests on

CAUTION

Trimble states, in identical words in two separate topics:

"Good RMS values do not mean that the calibration succeeded. A visual check is needed. On the other side, bad RMS values mean that the calibration failed." (TBC 24886, 25096)

A number can prove failure. A number cannot prove success. Only two things can suggest success: observations that took no part in the adjustment, and looking at the data (*Technical Manual §23*).

Everything in this section follows from that, and it is why QC here is **layered** rather than a single test. A residual measures how well an adjustment fitted the observations it was given. That is a narrower question than the one that matters.

16.2 The layers

Each layer catches something the others cannot. **None of them is optional because another was performed.**

#	LAYER	CATCHES	ARTEFACT
1	Field coverage verification (§10.2)	Missing passes, missing overlap	Field record
2	Intake checks (§12.2)	Transfer loss, wrong CRS, missing sensors	Intake record
3	Trajectory RMS review	Where the solution was weak — before any point cloud exists	Screen capture
4	Residuals on control	A broken adjustment	Targets pane
5	Residuals on independent check points	An adjustment that fits its own observations and is still wrong	Recorded manually
6	Visual inspection of the point cloud	Doubled surfaces, thickening at range, systematic tilt	No software artefact
7	Imagery inspection	Coverage gaps, exposure, blur, corrupted images	No software artefact

#	LAYER	CATCHES	ARTEFACT
8	Export-state confirmation (\$19.2)	Delivering the unregistered cloud	Screen capture

Two of the eight layers produce no software artefact at all. If a reviewer asks whether the visual check was performed and over what extent, the only possible answer is a record somebody wrote.

16.3 Trajectory RMS review

RECOMMENDED PRACTICE

The trajectory should be reviewed in RMS colouring before the point cloud is inspected.

It is available before any point cloud exists, it costs seconds, and it says where the solution degraded, for how long, and whether the degradation is at the ends of the mission (*Technical Manual §24*). That determines where to look in every later layer.

16.4 Residuals

Residuals on control points used in the adjustment are recorded, by component	
Residuals on independent check points are recorded, by component	These are the ones that mean something (\$7.3)
Which points were control and which were checks is recorded manually	TBC does not report it (<i>Technical Manual §22.7</i>)

NO PUBLISHED FIGURE

TBC 2025.21 states that signed residuals are "included in the report" without naming it, and the only mobile mapping report topic does not mention residuals. **Whether the residuals can be produced as a report, or must be transcribed by hand, determines how this record is kept.** Answerable in ten minutes with the software open.

16.5 Visual inspection

RECOMMENDED PRACTICE

A visual QC pass over a registered mission covers:

Check | Looking for

---|---

Overlapping passes in Cutting Plane View, dragged the full length | Doubled surfaces

Flat surfaces at range — a wall, a building face | Thickening with distance: attitude error or a calibration issue

The ends of the corridor | Where a Local adjustment stopped; where the smoother was weakest

The degraded stretches identified at layer 3 | Whether the registration actually fixed them

Vertical surfaces against horizontal | Systematic tilt

Features near control versus far from control | Residual growth with distance from constraint

IMPORTANT

Set the rendering to Scan Color for this. Without it, two offset surfaces read as one thick one and the defect the check exists to find is invisible (*Technical Manual §25*).

NO PUBLISHED FIGURE

The working cutting-plane thickness for these checks.

The constraint the decision has to satisfy. Mobile mapping error is **localised** — it arrives in stretches, not as scatter (*Technical Manual §3.2*). **An inspection strategy must therefore be capable of finding localised degradation, which means traversing the corridor rather than sampling it.** A sampling scheme that inspects ten places will find a problem that affects the whole job and miss the one that affects 300 m.

SET BY THE PROJECT

The corridor continuity inspection method and its coverage — how much of a corridor is inspected, and how that is decided.

16.6 Imagery inspection

RECOMMENDED PRACTICE

Check | Looking for

---|---

Coverage | Gaps where a camera stopped, or a run was not colorized

Exposure | Blown highlights, blocked shadows under canopy and in underpasses — both unrecoverable

Motion blur | Speed too high for the light available

Obstruction | Aerials, a following vehicle, a smear on the dome

Focus and contamination | Rain, dust, insects on the optical surface

Corrupted images | **These are silent** — a corrupted side camera image exports as black

Alignment with the point cloud | Colour in the wrong place at feature edges indicates a camera boresight issue

SET BY THE PROJECT

Whether the **file-size scan** for detecting silently corrupted imagery is used. It is a screening method proposed by this project and **not validated** (*Technical Manual §26*).

16.7 What QC does not do

TRIMBLE REQUIREMENT — THE VISUAL CHECK IS TRIMBLE'S INSTRUCTION, NOT OURS

Good residuals **shall not** be treated as removing the need for visual inspection. Trimble does not recommend the check; it states that one **is needed**, and says so in identical words in two topics (§16.1).

RECOMMENDED PRACTICE

More generally, **a QC layer should not be substituted by another**. Each catches something the others cannot, and which layers Parametrix requires is and.

16.8 Records this section requires

RECOMMENDED PRACTICE

For each registered mission:

Record | Source

---|---

Residuals on control points used, by component | Targets pane

Residuals on independent check points, by component | Same

Which points were control and which were checks | Manual — TBC does not report it

Run-to-run RMS statistics, if used | Results tab

Trajectory RMS picture | Screen capture

Visual check performed, by whom, covering what extent | No software artefact exists

Imagery check performed, by whom | No software artefact exists

Results of Scan Generation | §13.3

Mission Report | §18.3

17 Acceptance and Approval

17.1 What acceptance is

Acceptance is the point at which the data becomes the deliverable. It is a decision by a named person, recorded, that a dataset meets the project's stated accuracy requirement (§6.1).

Acceptance is always **against a requirement**. A dataset is not accepted because it looks good, or because the residuals are small, or because the schedule has run out.

17.2 There is no general numerical tolerance, and none is invented here

This SOP states no numerical acceptance tolerance. That is deliberate, and the reason matters more than the gap.

Trimble publishes no acceptance tolerance for the MX60	Its published figures are instrument performance under stated conditions , which is not a project acceptance criterion
Trimble publishes no attitude error budget for the point cloud	A useful range for a given tolerance therefore cannot be calculated from the documentation (<i>Technical Manual §16.5</i>)
A good RMS does not prove success	Trimble states it twice, in identical words. Only observations held out of the adjustment, and looking at the data, can suggest success (<i>Technical Manual §23</i>)

A number here would be worse than the gap. An invented tolerance would be quoted, relied on and eventually defended, and there would be nothing behind it. The gap is visible; a fabricated threshold would not be.

What this means in practice

The accuracy requirement comes from the project, in writing, before collection (§6.1). Acceptance is then a judgement against *that* requirement, supported by the evidence at §17.3 — independent check points, their residuals by component, and the visual inspection.

Whoever signs an accuracy statement signs on their own professional judgement, supported by the evidence in §16. That should be understood by the person signing and by whoever receives it.

What is not yet known about this system

Two relationships have not been measured on Parametrix's MX60 and cannot be read from the documentation:

- **How achieved accuracy varies with GNSS conditions.** Corridor geometry, canopy and urban reflection all degrade the trajectory, and by how much on this system is not established.
- **Which feature types are fit for horizontal or vertical control** at MX60 point density and incidence angle (§7.1).

Until they are measured, both are matters for judgement on the project, informed by the check evidence actually obtained on that project.

17.3 What acceptance requires

Five things are required before a dataset is accepted. These are structural and do not depend on a number.

#	REQUIREMENT
1	The project's accuracy requirement is stated in writing (§6.1)
2	Independent check points exist , were designated before registration, and took no part in any adjustment (§7.3)
3	Residuals on those check points are recorded , by component (§16.4)
4	The visual inspection was performed and recorded (§16.5)
5	Update Scans was run , and the accepted cloud is the registered one (§13.4)

Requirement 2 is the one that cannot be recovered afterwards. If no point was held out, the dataset cannot be checked later without re-surveying, because every point it might be checked against helped produce it.

17.4 What acceptance does not rest on

	WHY
RMS alone	TRIMBLE REQUIREMENT — a visual check is needed. Trimble says so twice, in identical words (§16.1)
Residuals on control points used in the adjustment	They measure the fit of the adjustment to observations it was given

WHY	
A clean-looking point cloud	Mobile mapping data does not look wrong when it is wrong (<i>Technical Manual §3.2</i>)
Agreement between two passes	Two passes can agree with each other and both be displaced, if the trajectory was drifting through the stretch

17.5 Who accepts

The Project Surveyor accepts a registration and signs an accuracy statement — and not the person who computed the registration (§4.2). Where one person must do both on a small job, the acceptance and its evidence are recorded so that the basis of the decision can be reviewed afterwards.

17.6 The accuracy statement

An accuracy statement issued to a client states:

The accuracy requirement it was assessed against	
The evidence — residuals on independent check points, by component	
The basis of the check — how many points, where, in which GNSS environments	
The extent it applies to	Accuracy is not uniform along a corridor (<i>Technical Manual §3</i>)
Any segment excluded, and why	§8.6, §22

Accuracy varies along the corridor. A single figure for a whole corridor implies a uniformity the method does not have. Where the statement is a single figure, the extent and the conditions it applies to are stated with it.

17.7 Records this section requires

RECORD
Acceptance decision — dataset, by whom, date, against what requirement

RECORD
The evidence the decision rested on
The accuracy statement issued

In Plain Language

Accepting the data is a separate act from producing it. Somebody puts their name to a statement that this dataset meets the accuracy the project asked for.

Why it matters. There is no company-wide number to check against, and no Trimble number either. The check is against what the project asked for in writing, using points that were deliberately kept out of the adjustment so they could still tell you something.

Remember this. You cannot check accuracy with points that helped produce the answer. If nothing was held back before registration, the dataset cannot be checked afterwards without going back out and surveying more.

If the steps are skipped. A dataset gets delivered with a number attached that nobody can support. The failure surfaces when a client or a reviewer asks how the figure was arrived at — usually long after the corridor has changed and re-surveying it is no longer cheap.

18 Destructive Operation Controls

18.1 What this section governs

An operation that removes data or history and cannot be undone within the software. In the MX60 workflow, one command is in this class.

CAUTION

Cleanup Mobile Mapping Mission is destructive and cannot be undone.

It permanently removes registrations, trajectories and scan sets from the project, keeping only the most recent. Trimble states: *"Please, have a backup copy of your project prior performing the operation, it cannot be undone."* (TBC 26466)

Clearing a data disk before a verified copy exists (§11.2) and deleting raw mission data (§21) are destructive in the same sense, and are governed by the same principle: **authorisation, and the record made before the act, not after.**

18.2 Authorisation

SET BY THE PROJECT

When may Cleanup be performed, by whom, and what must be archived first?

RECOMMENDED PRACTICE

Cleanup should not be run without written authorisation from the person with the authority under §4 — proposed there as the Project Surveyor. "In writing" carries the meaning in §3.3.

The verb is *should* because the authority does not exist yet. blocks this operation: until Parametrix names who may authorise a Cleanup, there is nobody for the clause to name.

18.3 What is archived first

RECOMMENDED PRACTICE

A sequence in which Cleanup destroys nothing that matters:

| Step | Why in this order

---|---|---

1 | **Complete and accept QC** (§16, §17) | Cleanup is an end-of-preparation step. Running it before acceptance removes the alternatives you might need to go back to

2 | **Run the Mission Report and archive it** | It records capture devices, runs, trajectories, generated scans, and per-sensor calibration with date (*TBC 23991_1, 24868*). **Run it before Cleanup** — afterwards it can only report what survives

3 | **Record the registration evidence** (§16.8) | Control/check designation and residuals. TBC does not appear to report these

4 | **Archive** `Targets.csv` (*TBC 22905*) | The picked registration observations — the registration's field book, and not recoverable

5 | **Archive the numbered SBET files** `sbet_<date>_reg_####.out` | Pending, assume Cleanup removes them

6 | **Archive the calibration JSON** (§15.5) | The system state the mission was processed under

7 | **Take the project backup Trimble asks for**, to a location that is part of the project archive (§21) | A backup nobody can find is not a backup. Not a local copy on the processor's machine

8 | **Obtain the authorisation** §18.2 requires |

9 | **Run Cleanup** |

10 | **Record that it was run** — by whom, on what date, and what was archived first | Otherwise the absence of history is itself unexplained

Steps 2 to 6 are small files. The whole set is a few megabytes beside a project of tens of gigabytes, and they are the difference between a deliverable that can account for itself and one that cannot.

NO PUBLISHED FIGURE

Does Cleanup delete `sbet_*_reg_####.out` **from storage, or only remove the project objects?** The answer determines what step 5 has to cover. Until it is known, assume the worse case.

18.4 Why this matters more here than elsewhere

Cleanup removes the evidence of how a deliverable was produced, and MX60 provenance is already the hardest part of this workflow (*Technical Manual §30*). A project that has been cleaned up can no longer show which of several trajectories a delivered cloud was built on, because the alternatives are gone along with the record of which one was chosen.

That is not an argument against running Cleanup. It is the argument for §18.3.

18.5 Records this section requires

- Authorisation — who, when, for which mission
- What was archived before, and where it is
- That Cleanup was run, by whom, on what date

19 Export and Delivery Controls

19.1 The release gate

Export is the last point at which a mistake is still internal. This section is a gate, not a procedure: the method is in the Office How To.

19.2 Before any export

RECOMMENDED PRACTICE

Before any export from a registered mission, confirm in Project Explorer that the scan nodes selected for export sit beneath the intended registered trajectory, and that their stations carry the `_reg_####` suffix.

CAUTION

Registration does not modify the point cloud until Update Scans is performed.

An operator can complete a registration, obtain good residuals, accept the result, and then export point cloud data that still reflects the pre-registration trajectory. **The export succeeds. The file is valid. The data is unregistered.**

SET BY THE PROJECT

Is the pre-export trajectory-node confirmation mandatory, and may export be performed without it? This is the single cheapest control in the procedure and the one that prevents the most expensive failure.

NO PUBLISHED FIGURE

The reliable export-state verification method for each export path, and what happens when a Point Cloud tab selection is drawn across scans belonging to two different trajectories. Neither is documented.

19.3 Export timestamps

CAUTION

With **Export timestamps** set to **Yes**, Trimble states that *"the exported scans are **reprocessed from the raw data**"* rather than being the scans processed with Generate Scans (TBC 23339, 22501).

Which trajectory that reprocessing uses is not stated. Until this is established, the exported data may not be the data that was registered and checked.

19.4 What the delivery carries

REQUIREMENT	
Grid or ground, as agreed (§6.3)	A ground-scaled export does not record the scale factor it used ; a grid export writes a sidecar
The coordinate reference system, datum and epoch	Stated in the delivery, not only in the file
The delivery record (§20)	
The accuracy statement (§17.6)	Where accuracy is relied on

SET BY THE PROJECT

Deliverable specification — standard formats, which export path produces each, and the default scaling.

19.5 Provenance limitation, stated plainly

IMPORTANT

Exported mobile mapping data may retain coordinate, timing, and in some formats trajectory information, but the captured Trimble documentation does not establish that the output uniquely identifies the adjusted trajectory or registration result used to create it.

That is why the delivery record in §20 exists, and why it is not optional.

NO PUBLISHED FIGURE

Which trajectory travels with a publish or an export; what a LAS file actually carries in its header, VLRs and sidecar; whether exported imagery reflects a registration; and whether a delivered dataset can be matched back to its trajectory after the fact.

19.6 Imagery

SET BY THE PROJECT

What is Parametrix's position on imagery privacy? Are unblurred originals retained, and for how long? Mobile mapping imagery captures faces, number plates and private property as a matter of course. **The decision is made before collection, not on request.**

19.7 Records this section requires

- Export-state confirmation, before export
- The delivery record — §20
- What was delivered, to whom, when, in what format and scaling

20

Documentation and Records

20.1 Why the record carries unusual weight here

In conventional survey work the observations are the record: field notes, raw files, an adjustment report. In mobile mapping, several of the things that determine whether a deliverable is correct **produce no software artefact at all**.

FACT	SOFTWARE ARTEFACT
Which points were control and which were independent checks	None — TBC does not report it (<i>Technical Manual §22.7</i>)
That the visual inspection was performed, and over what extent	None
That the imagery inspection was performed	None
The conditions at collection — occlusion, weather, traffic	None
Which trajectory a delivered cloud was built on	Partial, and not conclusive (<i>Technical Manual §30</i>)

Those five facts are either written down at the time or lost.

20.2 The record package

SET BY THE PROJECT

What provenance record accompanies a deliverable, where does it live, and who produces it?

RECOMMENDED PRACTICE

A minimal package that closes the gap:

| Artefact | Size

---|---|---

- 1 | **A one-page delivery record** per dataset: mission ID, trajectory node name, SBET filename including its `_reg_####` number, registration type, export path and date, exported by whom | kB
- 2 | **The control and check table** — point ID, Use XY, Use Z, As Check, and the residual on each | kB
- 3 | **The Mission Report**, run **before** Cleanup (§18.3) | kB
- 4 | **The calibration JSON** in force (§15.5) | kB
- 5 | `Targets.csv` (§18.3) | kB
- 6 | **The field record** (§9.6) | kB
- 7 | **The QC record** (§16.8), including the two layers with no software artefact | kB

Seven artefacts. Five of them are small files that already exist; two are written by a person. **The whole package is a few hundred kilobytes.**

20.3 Where records live

In the project record (§3.3) — durable, not on a processor's local machine, and findable by someone who was not involved.

SET BY THE PROJECT

Folder structure, naming and storage location (§11.4).

20.4 When they are made

At the time. A record reconstructed later is a reconstruction, and the facts most worth having are exactly the ones that cannot be reconstructed.

20.5 Index of records this procedure requires

The full index, section by section, is **Appendix A**.

20.6 What a record is for

Not compliance. Three specific uses, each of which has happened to somebody:

A question about a deliverable, years later	"Which trajectory was this built on, and what was it checked against?"
A test result that changes what a past deliverable means	and both have this shape. If either returns a result meaning a past export was not what it was believed to be, the record is what identifies which deliverables are affected
Improving the work	The relationship between GNSS conditions, control density and achieved accuracy on Parametrix corridors is not published anywhere and cannot be, because it is specific to this system and these roads. It can only be learned from a record of what was done and what came out

20.7 Records this section requires

- The delivery record, per §20.2
- The control and check table, with residuals
- Where the record package lives for a given project

21 Retention and Archive

21.1 What has to survive, and for how long

SET BY THE PROJECT

What is retained, where, for how long, and by whom?

The decision is genuinely a decision — storage cost against the ability to reprocess — but it divides cleanly, because the artefacts fall into three tiers of very different size.

21.2 The three tiers

RECOMMENDED PRACTICE

Tier 1 — small, irreplaceable, and the basis of any later defence

ARTEFACT	SIZE
Field record	kB
Control and check designation table, with residuals	kB
Mission Report, run before Cleanup	kB
Calibration JSON in force at processing	kB
Results of Scan Generation	kB
Delivery record (§20.2)	kB
QC record (§16.8)	kB
The accuracy statement issued to the client	kB

The whole of Tier 1 is a few hundred kilobytes. There is no storage argument against retaining it indefinitely, and every one of these is unrecoverable.

Tier 2 — retained for a defined period

ARTEFACT	SIZE
SBET, its processing report, and the frame and epoch log	MB
Numbered registered SBETs <code>sbet_*_reg_####.out</code>	MB
<code>Targets.csv</code>	kB
Base station data	MB
The delivered files	GB

Tier 3 — retained per the decision

ARTEFACT	SIZE	NOTE
Raw mission folder	Tens to hundreds of GB	The only thing that permits reprocessing. Once gone, the deliverable cannot be improved — only re-collected
TBC project	Tens to hundreds of GB	The provenance record (<i>Technical Manual §30</i>)

The Tier 3 decision is the expensive one and it is a real trade. Keeping raw data means a better trajectory later is possible — from better base data, a POSpac upgrade, or LiDAR QC. Discarding it means the deliverable is final in a way it need not have been.

21.3 What is not discarded on one person's judgement

RECOMMENDED PRACTICE

`POS_1/raw/` and `Targets.csv` should not be deleted by an individual acting alone.

Both are irreplaceable, and the reason is a fact about the data rather than a policy: one cannot be recomputed from anything, and the other was picked by a person and would be different if picked again (*Technical Manual §5.4*). **What the proposal needs from Parametrix is who else has to agree**, which is part of.

21.4 The archive record

RECOMMENDED PRACTICE

One page per project, in the archive, stating: what was archived, where, when and by whom; the retention tier applied; whether Cleanup was run and what was archived first; and where the raw mission data is, if it is held elsewhere.

Without it, the archive is a folder somebody has to reverse-engineer.

21.5 Records this section requires

- The archive record
- Retention tier applied, and the date the period runs from
- Disposal, where it occurs — what, when, authorised by whom

22 Non-conformance and Re-collection

22.1 What counts

A non-conformance is a departure from a **shall** in this procedure, or a dataset that does not meet the project's accuracy requirement (§17).

It is not a non-conformance to be unable to meet a requirement Parametrix has not set. Where a clause here is a **recommended practice** rather than a requirement, departing from it is a decision to record, not a failure to report.

22.2 Raise it, do not absorb it

Anyone who identifies a non-conformance raises it. The person who finds it is frequently the person who caused it, and the procedure is designed so that raising it is the normal thing to do.

A non-conformance that is fixed quietly leaves no trace that the workflow failed — which means it will happen again, to somebody else, on a job where it costs more.

22.3 The two categories

Recoverable in the office	Reprocess the trajectory, regenerate scans, re-register, re-export. Costs hours. Requires that the raw data was retained (§21)
Recoverable only in the field	Missing coverage, missing overlap, a mission with no closing sequence, a weak initialization. Costs a mobilisation

The second category is why §10 exists: **the field stage is the only stage with no office remedy** (*Technical Manual §5.5*).

22.4 Re-collection

SET BY THE PROJECT

Who decides that a dataset is rejected and that re-collection is required, and who bears it?

The decision is uncomfortable and is therefore the one most likely to be deferred until it is more expensive. Naming the decision-maker in advance is most of the control.

22.5 When a segment cannot be served

Where a corridor segment produces an unacceptable trajectory and no remedy applies, the honest finding is that **mobile mapping may not be the appropriate acquisition method for that segment**.

SET BY THE PROJECT

The decision rule: who decides, against what, and what the client is told. The planning-stage counterpart is §8.6 — identifying such segments before mobilising is much cheaper than finding them afterwards.

22.6 When a test result changes what a past deliverable means

Two open tests have this shape: (whether exporting with timestamps substitutes reprocessed data) and (which trajectory travels with an export or publish).

CAUTION

If either test returns a result meaning a past deliverable was not what it was believed to be, **that is not a documentation problem**. It is recorded in the Technical Manual and escalated.

The decision about what a client is told is not the tester's to make, and it is not specified here. Who it escalates to is part of.

22.7 Records this section requires

- The non-conformance — what, when, found by whom
- Disposition — accepted with qualification, reprocessed, re-collected, or rejected
- Where re-collection occurred, what changed
- Any deliverable affected, and what was done about it

A Index of Required Records

Five of these records have no software artefact behind them — the control and check designation, the visual inspection, the imagery inspection, the field conditions, and the disposition of a non-conformance. They are written by a person or they do not exist (§20.1).

A1 · By section

§	SECTION	RECORD
4	Roles, Responsibilities and Authorities	Who holds each role on a given project
		Who designated control versus independent check, and when
		Any exercise of stand-down authority, and its reason
5	Competence and Training	Who holds which qualification, and from when
		Training delivered, and against which document
6	Project Setup Requirements	Accuracy requirement, and its source
		CRS, datum, epoch, geoid — and who set them
		Grid or ground, as agreed
		Configuration assumed — Premium — and the fitment assumed with it
7	Control Requirements	Control network, with coordinates and their source
		Which points are control and which are independent checks, fixed before registration
		Who designated them, and when
8	Mission Planning Requirements	The mission plan — route, passes, direction, overlap
		GNSS assessment, with duration estimates and mitigations
		Initialization locations, primary and backup
		Segments mobile mapping will not serve, and what is proposed instead
9	Field Acquisition Requirements	Pre-flight confirmation, including aiding-sensor activation where fitted

§	SECTION	RECORD
		The field record, per §9.6
		Any exercise of stand-down authority, and its reason
		Operating-limit exceedance, if any, and what was done
10	Field Close-out and Handoff	Coverage verification, performed and by whom
		Deviations from the mission plan
		Handoff — what was transferred, to whom, when
11	Data Transfer and Custody	Transfer performed, verified how, by whom, when
		Location of the raw-data backup
		Custody, where required
12	Office Intake Requirements	Intake checks performed, by whom, with the result of each
		Calibration state at collection — <code>Extcal.json</code> and the Mission Report
		Any intake check that failed, and what was done
13	Processing Requirements	Trajectory processing settings, and the frame and epoch log
		Results of Scan Generation
14	Registration Requirements	Registration name, type, and the runs included
		The trajectory node produced, and its SBET filename with its <code>_reg_####</code> number
		Control and check designation, with the residual on each point, by component
		Confirmation that Update Scans was run
		<code>Targets.csv</code> , archived
		That the visual check was performed, by whom, over what extent
15	Calibration Control	Calibration performed — date, site, who, and the result including the visual check
		The calibration JSON, archived outside the project
		Which calibration each mission was processed against

§	SECTION	RECORD
		Periodic verification, when performed
16	Quality Control Requirements	Residuals on control points used, by component
		Residuals on independent check points, by component
		Which points were control and which were checks
		Run-to-run RMS statistics, if used
		Trajectory RMS picture
		Visual check performed, by whom, covering what extent
		Imagery check performed, by whom
		Results of Scan Generation
		Mission Report
17	Acceptance and Approval	Acceptance decision — dataset, by whom, date, against what requirement
		The evidence the decision rested on
		The accuracy statement issued
18	Destructive Operation Controls	Authorisation — who, when, for which mission
		What was archived before, and where it is
		That Cleanup was run, by whom, on what date
19	Export and Delivery Controls	Export-state confirmation, before export
		The delivery record — §20
		What was delivered, to whom, when, in what format and scaling
20	Documentation and Records	The delivery record, per §20.2
		The control and check table, with residuals
		Where the record package lives for a given project
21	Retention and Archive	The archive record
		Retention tier applied, and the date the period runs from

§	SECTION	RECORD
		Disposal, where it occurs — what, when, authorised by whom
22	Non-conformance and Re-collection	The non-conformance — what, when, found by whom
		Disposition — accepted with qualification, reprocessed, re-collected, or rejected
		Where re-collection occurred, what changed
		Any deliverable affected, and what was done about it

A3 · The records with no software artefact

These are the ones that get lost, because nothing in the software produces them and nothing complains when they are absent.

RECORD	SECTION	WHY NOTHING PRODUCES IT
Which points were control and which were independent checks	§7.3, §14.10, §16.4	TBC shows the state while the command is open and reloads it on Edit, but no report of it has been found (<i>Technical Manual §22.7</i>)
That the visual inspection was performed, and over what extent	§16.5	It is a human act in a viewer
That the imagery inspection was performed	§16.6	The same
Conditions at collection — occlusion, weather, traffic, what was not collected and why	§9.6	Nothing in the vehicle records them
Disposition of a non-conformance	§22.7	—

A4 · The smallest package that would satisfy the record

§20.2 proposes seven artefacts totalling a few hundred kilobytes. **Five of the seven already exist as files** and need only to be copied out of the project before it is cleaned up (§18.3). Two are written by a person.

